

Good afternoon.

My name is Parker McCrea from Waukee High School, and my project is Waukee Lost & Found, a fully coded website for schools to manage their lost and founds the right and most efficient way.

Let's be honest, most high schools lost and found are basically a cardboard box near the office. And that works... until it doesn't.

Last semester, I lost my graphing calculator right before a test. If you've ever bought one of those, you know they are not cheap. I checked the office, retraced my steps, and asked teachers, but nothing showed up. The biggest problem was not just that it was missing. The problem was that there was no real system.

No searchable database. No tracking process. No claim verification. No administrative review.

So I asked myself: what would a modern, secure, school-managed lost and found system look like if it were designed properly?

That is what I built.

Waukee Lost & Found is a database-driven web application that allows students to report lost items, submit found items with photos, search approved listings, and submit claims. This fully addresses the assigned topic. I also stuck with waukee's colors of purple, gold, black, and white to make it a familiar user experience, as well as having clear and readable fonts.

For development, I used PHP for the server side logic, phpMyAdmin to manage the database, apache netbeans as my IDE to code it, and MAMP to host it locally. I used html, css, and JavaScript as well. This was not built with templates. I coded it all from scratch, and I also separate controllers, models, and views using an MVC-style structure. Along with being separated in these folder categories, files are also separated by function to maintain readability and scalability.

On the user side, students can submit lost item reports or found item reports. Found item submissions can include photos, which makes it easier for users to identify items visually. The form also uses form validation, so users must fill out all required fields before submission. We all know how high schoolers can be, so along with this, I added a swear word blocker to prevent submissions that contain swear words or offensive language.

Approved found items are displayed in searchable listings, organized by most recent, so users do not have to dig through a physical box. When a user clicks on an item, they can view details such as the item name, location, description, image, and contact information if available. They can also file a claim.

The claim system is one of the most important parts of the project. Instead of allowing public instant claiming, the user has to submit a proof description. For example, if someone claims a calculator, they may need to describe a sticker, case color, or other detail that proves ownership.

This helps prevent false claims. Afterwards, if a claim is approved, a user gets an email that it has been approved as well as further follow up instructions. Same as if it was not approved.

On the admin side, staff can review found item submissions, approve or reject items, manage claims, and maintain control over what appears publicly. This makes the website more realistic for a school environment because it protects valuable items and creates accountability.

The database design is what holds all of the website's data. I created tables for lost items, found items, claims, users, locations, and logs. These tables hold information like emails, things submitted in the form, the timestamp of when it happened, etc.

Security was a major focus. The website uses password hashing to store users' passwords in our database with random letters, numbers, and symbols, prepared statements with PDO to protect against SQL injection, and role-based access so regular users cannot access admin pages.

Usability and accessibility were also important. Students come from all different backgrounds. I integrated accessibility features including ARIA labels, accessibility for screen readers and multi language compatibility, as well as using high contrast so it is easy to read. This also helped me stay ADA compliant. Here is an example of the screen reader. Now, here is an example of many different languages.

The website was built with Bootstrap 5.3 and tested on desktop, tablet, and mobile screen sizes to ensure full responsiveness. This matters because students, teachers, and admins may access the system from a phone, tablet, or school computer.

For resources, I used sources like Stack Overflow, bootstrap documentation, free code camp, and w3 schools to help me through the process of coding this.

So if I lost my graphing calculator again, I would not have to rely on a cardboard box or word of mouth. I could submit a lost item report in under a minute, check found items online, and know that if someone claims it, the process is verified and reviewed.

That is the difference between a manual process and my structured system.

Thank you. I would now like to demonstrate the website and its key functionalities.